
Deepening user involvement through Living Labs

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ABSTRACT

Over the last ten years there has been a development of the concept of living labs as quadruple helix innovation platforms, many with strong emphasis on ICT solutions in a variety of settings. In Europe, there has been some exploration of living labs in conjunction with HCI/UX approaches. In Australia, living labs are at an early stage of development. This paper reviews the state of development and intersection of living labs and HCI/UX approaches and the opportunities for living labs for practitioners in the Australian context.

CCS CONCEPTS

Human-centered computing~User centered design

KEYWORDS

Co-design, user-centred design, quadruple innovation helix, living lab approach

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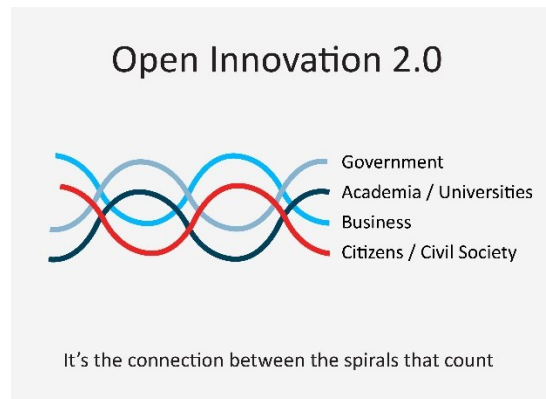


Figure 1: The quadruple helix model of open innovation 2.

1 INTRODUCTION

Innovation is a significant area of interest for governments, business and academia. Over the last 5 years there has been a shift in Europe [5] to put people at the heart of innovation processes. Businesses and governments started to embrace the concepts of open innovation, a term coined by Henry Chesbrough that advocates for organisation to incorporate external actors into their innovation processes. At policy levels the idea of triple helix innovation that brings together government, business and academia to innovate has given way to quadruple (and more) helix models. The quadruple helix model was first put forward by Carayannis and Campbell [1] who suggested the 'creative classes' represented the 4th helix (see Figure 1). In parallel to this Yawson [2] put forward the notions that the 4th helix is represented by civil society and citizens in general. It is this model that the European Union (EU) has promoted through its Open Innovation 2.0 policy [3]. This EU policy is a cornerstone of the Digital Single Market strategy for Europe, aiming to position themselves as a global digital technology leader [4], achieving this through the inclusion of citizens in the innovation process leading to innovation via faster cycles, better services and increased adoption.

"Open Innovation 2.0 (OI2) is a new paradigm based on a Quadruple Helix Model where government, industry, academia and civil participants work together to co-create the future and drive structural changes far beyond the scope of what any one organization or person could do alone. This model encompasses also user-oriented innovation models to take full advantage of ideas' cross-fertilisation leading to experimentation and prototyping in real world setting." [2].

In part, this policy was designed to address the European problem of high research quality and output but low translation into innovation; here innovation is being defined as what people actually adopt rather than what is created. The OECD suggests that 80% of the value of innovation is from adoption [5]. Hence, the value of innovation is not measured by its potential for change, but the actual value it creates to people's lives. With such a notion of innovation it is not surprising that users of innovative products become central to their conceptualisation, development and evaluation. This is very much in alignment with HCI approaches who have the inclusion of the user at the heart of their processes - however at a smaller scale and not always with the same aspirations of actual long-term impact. Living labs are the leading innovation platform [6] for quadruple helix innovation and we see opportunities for the Living Lab concept to be applied by HCI practitioners. In a European study of policies to support innovation, living labs were listed in the top three for innovation platforms and the only one to include end-users and real-life contexts.

The European Network of Living Labs (ENoLL) defines living labs platform as a "user-centred, open innovation ecosystems based on systematic user co-creation approach, integrating research and innovation processes in real life communities and settings" and emphasises that living labs operate as "intermediaries among citizens, research organisations, companies, cities and regions for joint value co-creation, rapid prototyping or validation to scale up innovation and businesses." [8].

At the *organizational* level living labs bring together different actors around a particular strategic area of interest, this could be a city or that of an aged care center. The lab manages the stakeholder relationships, builds the end-users community, and often provides supporting infrastructure. The living lab also manages access to the real-life settings where the interventions may take place. An example might be a city hosting a living lab in say a shopping mall and providing the support mechanisms for stakeholders innovate for the benefit of its citizens.

At the *program* level living labs will focus on systemic issues, looking at many interventions that might work together and build on each other over time rather than the solution to a single problem.

At the *project* level living labs use multi-methods approaches. There is no one single living lab methodology and each living lab will have its own specialisations. Research is integrated throughout the process. The overall approach brings a greater emphasis on user interaction and co-creation throughout the whole process. There are patterns that are common across living labs projects that include the use of iterative approaches, low fidelity prototypes, multi-disciplinary teams and co-creative methods. A living lab usually includes that the end user groups inform the area of research and need, their involvement in ideations, design and prototyping processes. This is much greater involvement than a test bed or similar approach where the interaction is much later and less consistent. Finally, the project is about delivering an innovation that will be adopted and improves the life of its community as well as serving economic interests.

The benefits of these projects are in the creation of value between the different stakeholders, business and governments through great adoption based on innovation and research. Co-creation of value between stakeholders in a living lab is central to the projects in a living lab setting.

The initial of European Living Labs concept emphasised the use development of ICT and some of the early handbooks [9] was human centred ICT based. Hence most labs have an ICT technologies component. The handbook and other literature from ENoLL and the Living Labs community emphasise strong user involvement in the innovation process, from ideation through to final user testing; users are treated as knowledgeable experts in their domain of experience and needs. Hence, by their nature living labs have a strong interest in user experience and for most labs the human computer interface.

The following methods are explicitly used in many living labs.

Coorevits et al [10] describe the use of personas in Living Labs. A persona approach here improved user experience and revealed unknown user needs through the continual updating of the personas through an open innovation process in a Living Lab setting. Rather than using the personas at the initial design setting the personas were updated throughout the process and enriched with data from the people the personas were based on as they imagined or used the various version in real life contexts.

Vandecasteele et al [12] and Vandenbroucke [11] show how large-scale user involvement in Living Lab approaches enabled the collection of significant UX data based on real user in context usage for a TV shopping app. The collection of detailed data facilitated an understanding of the different types of users, their emotional responses for use and non-use of the system. The overall pattern suggested that at the end of the process the number of users likely to adopt the system was higher than models predicted by Rogers innovation adopter's classification. Molinari [13] describes the use of living lab methods to create and understand holistic UX interactions for service co-creation in IoT environments. Pallot and Pawar [14] similarly describe the utility of Living Labs to explore and enhance UX approaches.

2 CHALLENGES AND OPPORTUNITIES

Whilst there are clear benefits to this approach there are challenges running a quadruple open innovation system in a traditionally triple helix world – in particular in regards to user involvement.

Stakeholders understand the benefit of having users involved, however, the typical expectation is to include users towards the end of the innovation development cycle – at the testing stage. In the start-up phase there is significant emotional attachment to the founder's idea. In other living lab contexts early involvement has been shown to lead to a high rate of pivoting and refining, stopping products that no-one wanted (saving capital expense) and major redevelopment of the concept. The current mindset in Australia tends to see the end-user group as a resource that value can be 'extracted' from rather than a mechanism to co-create value for the mutual benefit of both parties.

Intellectual property is another challenging area. Studies have shown the more open stakeholders can be the more likely new opportunities for value creation are to be found and the greater the success of the Living Lab. This requires a degree of trust between the different stakeholders.

The Future Self Living Labs is one example of an Australian living lab and was accepted into ENoLL in 2016. The Future Self Living Lab (FSDLL) has core development capabilities in innovative socio-technical systems and design solutions for maintaining and improving wellbeing and quality of life. The FSDLL develops technology, products and services with a focus on older adults. With its core and founding members the FSDLL is slowly building this trust with different parties sharing aspects of their business, research and project information. Here we described the experiences and challenges we had over the last two years in applying a living lab approach in practice – in particular promoting design-led innovation through co-creation with users.

IP and competition versus openness: Industry partners who are attracted to the living lab for temporary projects are, maybe understandably, less inclined to be open with their intellectual property. The challenge for the living lab is to find ways that temporary participants can contribute to the whole ecosystem, or preferably find ways to incorporate them as long-term participants who can develop a deeper relationship with other stakeholders enabling new opportunities for innovation.

In such a networked structure the living lab can find that there are participating partners who are competing in the same market. This may be businesses and civil society actors. For the lab this could present issues of conflicting interests. It also provides an opportunity for actors to collaborate rather than compete in every case, pooling resources and providing a better outcome for the end users.

Preset expectations versus open innovation. One of the key elements of the Living Lab model is discovery of unmet needs. Whether the living lab is addressing ageing or smart cities there is going to be a diversity of needs and approaches to resolve them. Living labs have emergent properties both in the sense of understanding needs and how those change with each intervention. A social practice maybe better than a technology service, or several services and social interactions may be needed. This runs counter to the way organisations, business, government and civil society usually attempt to tackle problems. Often there are preconceived ideas such as “we need an app to...”, or funding methods that dictate the innovation and outcome rather than focusing on an outcome with maximal long-term impact. With time there is a great opportunity to harness these emergent properties to discover and address those needs, in a variety of ways that meet the need of our citizens. Focusing on needs and outcomes removes the pressure to be disruptive or radical. One of our projects focused on personal alarms, which typically have a low compliance rate, leading to costly visits and checks by providers. A study in one project found that the functional aspects of the alarms were a good fit and there was little need for radical innovation from this perspective. However, we did find that the alarms did not meet the emotional goals of older adults; they found them stigmatising and impersonal, lacking genuine care. Redesigning these to meet emotional goals of being cared about and being unobtrusive we developed personal alarms that promoted interaction and wearable items that appeared more like jewellery.

Continuous user integration. The FSDLL currently has a user reference group that is intended to feed in areas of need and possible research directions into the living lab. This is still in its early stages with the intention of building the reference group and a larger group of participants that can consistently participate in living lab projects. Based on the experience of other labs there is significant potential in this approach, both for opportunity discovery and a great number of participants who can help refine services and version stages of development. With our community we also must recognise that group members pass away, need to go to hospital or have problems that do not represent the needs of a wider group.

The challenge of scaling. A great opportunity and challenge for a living lab is to scale products and take them to global markets. As living labs are context specific there has to be a careful selection of innovations that can be generalised outside of that local setting (e.g. a Melbourne based aged care centre) into a global version of the setting (e.g. aged care setting in Spain). Also limitations to scaling and hence for living labs to grow beyond their current size are most of the above described challenges concerning IP and preset expectations. However, the reference group and our local and international partners provide a great platform to help scale the right innovations from their local roots into global markets. As an ENoLL member we can also collaborate with other members to jointly run programs in the equivalent settings in multiple countries as transnational living lab. This has already been achieved in the context of Smart Cities.

3 CONCLUSION

There is a natural synergy between HCI/UX approaches and Living Lab methods with a strong focus on user involvement. HCI and UX are already incorporated into Living Labs in Europe. In Australia we also see great potential for Living Labs as a complementary innovation mechanism alongside many government and university initiatives. To make the most of a quadruple helix innovation model we need to educate stakeholders and manage their expectations to craft the conditions to create long term value between them and ultimately for the end user.

REFERENCES

- [1] Elias G. Carayannis, David F.J. Campbell. 'Mode 3' and 'Quadruple Helix': toward a 21st century fractal innovation ecosystem. *International Journal of Technology Management (IJTM)*, Vol. 46, No. 3/4, 2009
- [2] Robert M. Yawson. The ecological system of innovation: A new architectural framework for a functional evidence-based platform for science and innovation policy. 2009 University of Minnesota. The Future of Innovation: Proceedings of the XX ISPIM 2009 Conference, Vienna, Austria, June 21-24, 2009 https://mpra.ub.uni-muenchen.de/33179/1/MPRA_paper_33179.pdf
- [3] Digital Single Market, Open Innovation 2.0 published 4 December 2013 <https://ec.europa.eu/digital-single-market/en/policies/open-innovation>
- [4] Digital Single Market <https://ec.europa.eu/digital-single-market/en/open-innovation-20>
- [5] The Dublin Innovation Declaration, p14 23/05/2013 <https://ec.europa.eu/digital-single-market/en/news/%E2%80%9C-dublin-innovation-declaration%E2%80%9D-manifesto-ten-point-declaration-create-more-wealth-better>
- [6] Open Innovation Year Book 2016 31/05/2016, Open Innovation Strategy and Policy Group's (OISPG) <https://ec.europa.eu/digital-single-market/en/news/open-innovation-20-yearbook-edition-2016>
- [7] ENoLL About us <http://www.openlivinglabs.eu/aboutus>
- [8] ENoLL What are Living Labs <http://www.openlivinglabs.eu/node/1429>
- [9] The Living Lab Methodology Handbook Anna Ståhlbröst and Marita Holst, Social Informatics at Luleå University of Technology and CDT – Centre for Distance-spanning Technology, Sweden, 2012 https://www.ltu.se/cms_fs/1.1015551/file/LivingLabsMethodologyBook_web.pdf
- [10] Lynn Coorevits, Dimitri Schuurman, Kathy Oelbrandt, Sara Logghe. Bringing personas to Life: user experience design through interactive coupled open innovation. *Persona Studies* 2016
- [11] Karel Vanderbroucke, Dimitri Schuurman. Apptvate: The mobile shopping technology that enriches your tv-experience, iMinds-MICT- Ghent University.
- [12] Florian Vandecasteele, Karel Vandenbroucke, Dimitri Schuurman, Steven Vernstockt. Spott: On-the-Spot e-Commerce for Television Using Deep Learning-Based Video Analysis techniques. *ACM Trans. Multi media Comput. Commun. Appl.* 13 3s, Article 38 (June 2017), 16 Pages. <http://dx.doi.org/10.1145/3092834>
- [13] Francesco Molinari User Experience Analysis in Service co-creation: A Living Lab Approach. 2012. Service Design 2012 Third Nordics Conference on Service Design and Service Innovation
- [14] Marc Pallot & Kulwant Pawar. A holistic model of user experience for Living Lab experiential design, June 2012. DOI:10.1109/ICE.2012.
- [15] H., Chesbrough. Open Innovation: The New Imperative for Creating and Profiting from Technology By: W. Harvard Business School Press Books. Jan2003, p1. 272p. , Database: Business Source Complete